

Pointer Lab
C++ Programming
Instructor:
Dr Victor Govindaswamy
(100 points)



"Does running late count as exercise?" 😁



1

Cultivating Today, the Innovators of Tomorrow

Educating Today, the Geniuses of Tomorrow

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Design and Implementation of the CharCraft – C-String Workshop (3CS)



Objective

- The goal of this lab is to introduce C-style string manipulation using char* and the C string-handling library <cstring>. Students will practice working with character arrays, pointers, dynamic memory allocation, and menu-driven program design.



Learning Outcomes

- Use char* and dynamically allocated memory to store strings
- Apply common C string-handling functions correctly
- Design and implement a menu-driven C++ program
- Organize a program using classes and multiple source files
- Safely allocate and deallocate memory using new[] and delete[]
- Create a basic UML class diagram



Lab Description

You will create a C++ console application called CharCraft – C-String Workshop. The application allows a user to perform several string operations using CSTRING library functions only.

Important Notes:

- You must use char* for string storage.
- std::string may only be used for input convenience.
- All string operations must use <cstring> functions.



Required Files

You must submit exactly the following files:

1. StringManipulation.h
2. StringManipulation.cpp
3. StringManipulationTest.cpp
4. Makefile

Menu Requirements

Your program must display the following menu:

1) Compare two strings
2) Compare up to N characters
3) Copy a string
4) Append a string
5) Find the length of a string
6) Tokenize and reverse word order
0) Exit

Required String Operations

You must implement the following using <cstring> functions:

- strcmp
- strncmp
- strcpy
- strcat
- strlen
- strtok

Memory Management Rules

- All strings must be dynamically allocated using new[]
- All allocated memory must be released using delete[]
- Memory leaks will result in point deductions

Makefile Requirement (Beginner Level)

You must include a simple Makefile to compile your program.

Required Makefile content:

```
CharCraft: StringManipulation.cpp StringManipulationTest.cpp
	g++ StringManipulation.cpp StringManipulationTest.cpp -o CharCraft

clean:
	rm -f CharCraft
```

Usage:
make
./CharCraft
make clean

UML Requirement

You must submit a UML class diagram for the StringManipulation class showing:

- Class name
- Public member functions
- Private helper functions (if any)

Compilation Requirement

Your program must compile and run successfully on Linux using:
g++ StringManipulation.cpp StringManipulationTest.cpp -o CharCraft

Submission Instructions

Submit the following to Blackboard:

- All source files (.h and .cpp)
- Makefile
- UML diagram
- Script file showing program execution

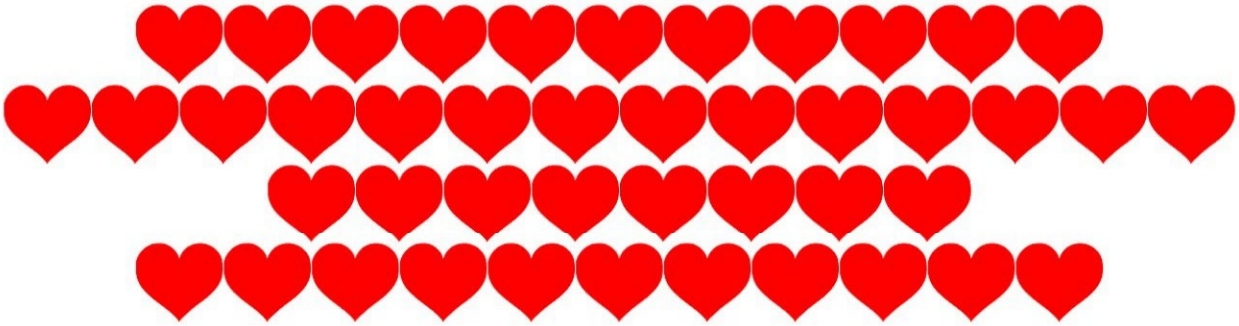
Grading Rubric (100 Points)

The Lab Quiz will be accessed based on:

Functionality (60 points)
Program Design (20 points)
Memory Management (10 points)
UML and Documentation (10 points)

Academic Integrity

This is an individual assignment. You may discuss concepts with classmates, but all code must be written by you.



Happy Coding!

